Improving AI Forecasting Performance Using Prompting and Judgmental Forecasting Techniques

Juliette Goldstein
The Chapin School
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ABSTRACT

This study evaluated whether prompting and judgment-based prediction techniques can improve the forecasting accuracy of large language models (LLMs), specifically ChatGPT (GPT-5.2). Using prediction market questions in the form of two datasets from Kalshi, six AI forecasting methods were tested: Scenario Forecasting, Prompt Engineering, Most Popular Option, Delphi, Ensemble, and a Meta model trained on prior forecasting performance, along with a Control model and a Human benchmark based on market open interest. Scenario Forecasting and Prompt Engineering were the only models to outperform the Control model, achieving accuracies of 58.6% and 58.3% on their predictions of the larger dataset. Forecasting accuracy declined sharply for the smaller dataset across all models and Human forecasts. In all, the data suggest that aggregation methods and exposure to existing predictions may reinforce counterproductive bias, yet structured reasoning prompts improve AI forecasting in data-rich environments.

INTRODUCTION

Although commonly associated with the weather, forecasting encompasses the general practice of making informed predictions about uncertain future events; it is central to

decision-making in politics, economics, healthcare, finance, and national security. Forecasting spans four primary domains: qualitative forecasting based on expert judgment, quantitative forecasting using numerical data, time-series modeling of historical trends, and causal forecasting based on historical cause-and-effect relationships [1]. In many applied settings, especially those involving rare or novel events, historical data may be limited or unavailable. This makes traditional machine learning (ML) techniques difficult to implement, as they typically must be trained on historical data. Lack of data can be addressed in several ways. In machine learning, specialized methods at each stage of the pipeline can help generate estimates; however, in most novel event scenarios, judgmental forecasting, the process of forecasting based on human intuition, is a necessity [2]. Indeed, it is widely recognized that "without management judgement in forecasting, serious problems can result," as illustrated by Nike's $400 million forecasting software failure attributed to insufficient human judgment [3].

Given the importance of human judgment in these settings, decades of research have focused on developing structured methods to improve judgmental forecasting. Strategies that are seemingly simple yet effective in uncertain forecasting domains can involve **scenario forecasting**, in which an expert imagines a wide variety of outcomes for a given event [4]. Aggregation techniques such as **wisdom of the crowd** reduce random and individual-specific error by averaging across multiple individuals' estimates, allowing systematic and random biases to partially cancel out. The **Delphi** method further structures this type of aggregation by collecting forecasts from a panel of experts anonymously over multiple rounds, revealing each previous round's forecasts between rounds to move toward a consensus [5]. **Judgmental bootstrapping** codifies expert reasoning by statistically modeling the implicit decision rules experts use, converting their judgment to predictive equations; these bootstrapped, or self-aided,

models can counterintuitively exceed the performance of the experts they are trained on [6]. **Dialectical bootstrapping** requires forecasters to generate an alternative perspective after forming an initial estimate, then average the two predictions [7]. This process can mitigate overconfidence—where one's subjective bias exceeds the objective truth—and anchoring bias—where one relies too heavily on an initial piece of information. Finally, an **ensemble** strategy can combine outputs from multiple forecasting models or techniques, often outperforming any single approach [8]. Together, these techniques demonstrate that aggregation of forecasts and structural aids to judgment can meaningfully improve predictive accuracy, particularly in environments with limited historical data.

At the same time, large language models (LLMs) have shown promise in predictive tasks, especially when supported by guidance in the form of **pre-prompting**, which is the process of assigning contextual information, instructions, and/or personas to an LLM before asking it to complete a task [9]. For example, recent research suggests that AI performance improves substantially when provided with instructions on how to structure reasoning [8]. This illustrates the need for strategies like ART (Automatic Reasoning and Tool-use), which breaks problems down into multiple steps, addressing the fact that LLMs struggle with multi-step reasoning and using external tools for specialized tasks like calculations, database lookups, or domain-specific queries [10]. Moreover, in one study, exposure to the **median human forecast** enhanced model accuracy by 17–28%, and averaging of human and machine predictions performed even better [8]. Finally, in **supervised learning**, a model can be trained on a dataset in which each input is paired with the correct output (e.g., input: image of cat, output: "cat"), allowing it to learn patterns that map inputs to predictions [11].

Building on this literature, this study investigates whether forecasts made by artificial intelligence can be systematically improved by incorporating methods that have been used to enhance human and artificial judgment. Specifically, Prompt Engineering (pre-prompts designed to structure reasoning), Scenario Forecasting, automated Delphi, referencing the modal human forecast (henceforth Most Popular Option), ensemble combination, and meta-training on prior forecasting performance (henceforth Meta) were evaluated. Using real prediction market questions from Kalshi, AI forecasts were compared with each other and with a Human forecast condition to determine which methods yield superior predictive accuracy [12]. By experimentally adapting human forecasting and AI prompting techniques to AI systems, this work contributes to a growing literature exploring how humans can leverage AI to become better at predicting the future.

MATERIALS AND METHODS

All forecasting questions were drawn from Kalshi, a prediction market that allows users to bet on the outcome of real-world events (like elections, sports, or economic data). Kalshi was selected for this study because of its popularity, plethora of events (forecasting questions), as well as its publicly-accessible and well-documented API (Application Programming Interface). An API allows a user to access resources from an application via code rather than the application's user interface itself. Each event and its corresponding markets (answer options) were retrieved using Kalshi's public API (https://api.elections.kalshi.com/trade-api/v2), which provides access to events available on Kalshi's website. While the site is designed for human browsing and trading, the API allows researchers to download the same market information in a machine-readable format for automated analysis and manipulation. Code and sample CSV files

from this study can be found on GitHub (https://github.com/JulietteGoldstein/AI-Forecasting-Using-Judgmental-Forecasting-and-Prompting-Methods).

Data was collected in two waves. In Wave 1, 700 events scheduled to resolve between December 31, 2025 and January 1, 2026 (EOY (end-of-year)) and 65 events scheduled to resolve between January 15, 2026 and January 24, 2026 (EOJ (end-of-January)) were collected. Forecasts were generated between December 14, 2025 and December 16, 2025. All artificial intelligence forecasts were generated using ChatGPT (GPT-5.2) through OpenAI's API. Six forecasting methods were evaluated, each implemented as a separate model. Each model was provided with the event description and the complete set of answer options for each given event. The **Scenario Forecasting** AI model was prompted to generate an optimistic (best-case), pessimistic (worst-case), and typical scenario before assigning probabilities to each and selecting the most likely answer. All prompts given to AI models can be found on this study's GitHub repository. In the **Prompt Engineering** model, a primary AI instance was given information on how providing structured directions to guide thinking can improve AI output. It was then asked to construct an instruction prompt that would guide a second AI session through its prediction; the second session's prediction was taken as the final prediction (see Figure 1). In the **Most Popular Option** model, the answer option with the highest open interest on Kalshi at the time of prediction was selected. Open interest represents the number of active contracts held by traders and reflects the amount of capital committed to a particular market, serving as a type of modal human forecast. The mode was used rather than the median because the vast majority of Kalshi's markets are qualitative. The AI model was then asked to make a prediction, having been given the human most popular option. The **Control** model was simply instructed to select an answer

option without any additional guidance. In the **Delphi** model, a primary AI instance generated five expert personas relevant to the question, each of which was assigned to a separate AI instance, which independently produced a forecast. After viewing one another's responses, the expert personas made revised predictions. This process was repeated until the personas had made four guesses. The most common final answer was selected from the fourth round of prediction. Finally, in the **Ensemble** condition, the model was shown the predictions from all other methods, then asked to select an answer option. Each Wave 1 model made predictions for both EOY and EOJ events.

Figure 1 Example excerpt of preprompt given to the Prompt Engineering model. PROMPT_ENGINEERING_NOTES contains notes on why prompt engineering is successful with AI and how to make a beneficial structured prompt.

```
You are an expert prompt engineer and forecasting methodologist.

Below are some notes about prompt engineering, forecasting, scenario reasoning,
human–AI ensembles, and tool-augmented reasoning:

{PROMPT_ENGINEERING_NOTES}

Design an instruction prompt that will be given to another AI model to make a probabilistic forecast and then
select ONE answer option.

  Requirements for the engineered instruction prompt:
  1. The specific listed instructions should:
     - Decompose the question-specific reasoning process into smaller logical steps.
     - Use outside real-world knowledge, but DO NOT actually call tools or APIs.
     - Not be vague; they should be specific to the prompt at hand.
  2. It should warn about prompt brittleness and emphasize following the format strictly.
  3. It must specify final visible output format:
     <exactly ONE of the provided answer options, copied verbatim, with no additional words or labels>
  4. The engineered prompt should NOT answer the question itself.
```

In Wave 2, forecasts were generated on January 3, 2026 for all EOJ events. This wave served as short term predictions to compare with the long term Wave 1 EOJ predictions. Because of the smaller sample size (n = 65), each forecasting method was repeated three times to reduce variance. Wave 2 included all methods used in Wave 1 as well as a Meta model trained on Wave

1 performance. The Meta model was inspired by the concepts of supervised and meta-learning. Meta-learning, or "learning to learn," is a machine learning paradigm that can train an algorithm across a wide range of tasks to create generalizable strategies, enabling the model to adapt rapidly to new tasks with minimal data in the future [13]. To construct the Meta model, the actual resolved outcome for each Wave 1 question was identified using Kalshi's official settlement result. For each Wave 1 model, accuracy was computed as the proportion of correct predictions to total predictions. Laplace smoothing implemented as Bayesian shrinkage was used to smooth these accuracy values by shrinking them toward 50%. Inspired by the posterior mean estimator of the Beta–Binomial model described by Gelman et al. in *Bayesian Data Analysis*, probability estimates were smoothed to reduce the influence of small sample sizes. The equation is given by [14]:

$$E[\theta|y] = \frac{\alpha + y}{\alpha + \beta + n}$$

where θ is the true probability of success, y is the number of observed successes, n is the sample size, and α and β are prior pseudocounts for success and failure. The derivation is as follows:

Let $y = \hat{p} \cdot n$

where $\hat{p}$ is the observed accuracy, and n is the number of observations.

$$\tilde{p} = \frac{\hat{p} \cdot n + \alpha}{n + \alpha + \beta}$$

where $\tilde{p}$ is the smoothed accuracy and $\hat{p} \cdot n$ is the observed number of correct predictions.

$$\tilde{p} = \frac{\hat{p} \cdot n + k/2}{n + k/2 + k/2}$$

where $k = \alpha + \beta$ is the total prior strength, so setting $\alpha = \beta = k/2$ shrinks the estimate toward 0.5, the chance level for a binary outcome. Thus,

$$\tilde{p} = \frac{\hat{p} \cdot n + 0.5 \cdot k}{n + k}$$

where $\tilde{p}$ represents smoothed accuracy, $\hat{p}$ represents raw accuracy estimate, k represents the number of pseudo-observations at 50% accuracy, and n represents sample size. This equation assumes that each model's true accuracy is uncertain and combines the observed data with a small number of neutral "pseudo-observation" data points at 50% accuracy (in this case, k = 10) [15]. This formula essentially prevents the model from being overconfident when it only has a small amount of data to look at. For each event, the Meta model was provided with these smoothed accuracy estimates, the question, the answer options, and the Wave 1 models' predictions of that event. Based on this information, the Meta model was instructed to select one answer option.

Because the bulk of this procedure was submitted to supervisors on October 21, 2025, the methodology of this study was effectively preregistered. Prior to analysis, duplicate questions, markets without resolutions, and markets with zero open interest at the time of prediction were excluded. Each prediction was scored as either correct or incorrect by comparing the model's selected answer option to Kalshi's officially settled market outcome. Moreover, the three Wave 2 repetitions were reduced to one repetition by majority vote, so as not to weigh the samples so heavily. After outliers were removed, the EOY dataset contained 690 events and the EOJ consisted of 65.

Accuracy comparisons were conducted using both paired and unpaired two-tailed statistical tests. McNemar's exact test was used for paired binary comparisons, appropriate when

evaluating two forecasting methods on the same set of questions. This test assesses whether the number of cases in which one method is correct and the other incorrect differs significantly from the reverse pattern. Comparisons included each model versus the Human benchmark (most popular open interest at the time of prediction), each model versus the Control model, and Wave 1 models versus their Wave 2 counterparts on overlapping question sets. For comparisons involving unequal sample sizes across EOY and EOJ events, two-proportion z-tests were used to assess differences in independent accuracy proportions. Statistical significance was evaluated with a threshold of $p < 0.05$.

RESULTS

Table 1 Accuracy scores of all models across all waves and datasets. Notably, Scenario Forecasting and Prompt Engineering have the highest accuracies overall.

Wave	Dataset	Forecaster	Events Forecasted (n)	Total Correct Predictions	Accuracy (%)
1	EOY	Humans	690	122	17.7
1	EOY	Scenario Forecasting	690	404	58.6
1	EOY	Prompt Engineering	690	402	58.3
1	EOY	Most Popular Option	690	195	28.3
1	EOY	Control	690	375	54.3
1	EOY	Delphi	690	331	48.0
1	EOY	Ensemble	690	353	51.2
1	EOJ	Humans	65	9	13.8
1	EOJ	Scenario Forecasting	65	8	12.3
1	EOJ	Prompt Engineering	65	7	10.8
1	EOJ	Most Popular Option	65	10	15.4
1	EOJ	Control	65	7	10.8
1	EOJ	Delphi	65	9	13.8
1	EOJ	Ensemble	65	8	12.3
2	EOJ (majority vote)	Humans	65	9	13.8
2	EOJ (majority vote)	Scenario Forecasting	65	7	10.8
2	EOJ (majority vote)	Prompt Engineering	65	6	9.2
2	EOJ (majority vote)	Most Popular Option	65	9	13.8
2	EOJ (majority vote)	Control	65	8	12.3
2	EOJ (majority vote)	Delphi	65	10	15.4
2	EOJ (majority vote)	Ensemble	65	9	13.8
2	EOJ (majority vote)	Meta	65	8	12.3

Table 2 McNemar tests comparing all model predictions to Human forecasts across all waves and datasets show that Wave 1 EOY predictions were significantly more accurate than Human predictions.

Wave	Dataset	Comparison	Events Forecasted (n)	P-value	Significant ($p < 0.05$)	Higher Accuracy
1	EOY	Humans vs Scenario Forecasting	690	6.97×10^{-59}	TRUE	Scenario Forecasting
1	EOY	Humans vs Prompt Engineering	690	2.66×10^{-59}	TRUE	Prompt Engineering
1	EOY	Humans vs Most Popular Option	690	2.29×10^{-14}	TRUE	Most Popular Option
1	EOY	Humans vs Control	690	1.85×10^{-51}	TRUE	Control
1	EOY	Humans vs Delphi	690	2.21×10^{-39}	TRUE	Delphi
1	EOY	Humans vs Ensemble	690	3.76×10^{-47}	TRUE	Ensemble
1	EOJ	Humans vs Scenario Forecasting	65	1	FALSE	—
1	EOJ	Humans vs Prompt Engineering	65	0.754	FALSE	—
1	EOJ	Humans vs Most Popular Option	65	1	FALSE	—
1	EOJ	Humans vs Control	65	0.791	FALSE	—
1	EOJ	Humans vs Delphi	65	1	FALSE	—
1	EOJ	Humans vs Ensemble	65	1	FALSE	—
2	EOJ (Majority Vote)	Humans vs Scenario Forecasting	65	0.754	FALSE	—
2	EOJ (Majority Vote)	Humans vs Prompt Engineering	65	0.549	FALSE	—
2	EOJ (Majority Vote)	Humans vs Most Popular Option	65	1	FALSE	—
2	EOJ (Majority Vote)	Humans vs Control	65	1	FALSE	—
2	EOJ (Majority Vote)	Humans vs Delphi	65	1	FALSE	—
2	EOJ (Majority Vote)	Humans vs Ensemble	65	1	FALSE	—
2	EOJ (Majority Vote)	Humans vs Meta	65	1	FALSE	—

Table 3 McNemar tests comparing all model predictions to the Control across all waves and datasets show that EOY Scenario Forecasting and Prompt Engineering predictions were significantly more accurate than Control predictions.

Wave	Dataset	Comparison	Sample Size (n)	P-value	Significant ($p < 0.05$)	Higher Accuracy
1	EOY	Humans vs Control	690	1.85×10^{-51}	TRUE	Control
1	EOY	Scenario Forecasting vs Control	690	0.005	TRUE	Scenario Forecasting
1	EOY	Prompt Engineering vs Control	690	0.021	TRUE	Prompt Engineering
1	EOY	Most Popular Option vs Control	690	4.28×10^{-33}	TRUE	Control
1	EOY	Delphi vs Control	690	5.40×10^{-5}	TRUE	Control
1	EOY	Ensemble vs Control	690	0.013	TRUE	Control
1	EOJ	Humans vs Control	65	0.791	FALSE	—
1	EOJ	Scenario Forecasting vs Control	65	1	FALSE	—
1	EOJ	Prompt Engineering vs Control	65	1	FALSE	—
1	EOJ	Most Popular Option vs Control	65	0.549	FALSE	—
1	EOJ	Delphi vs Control	65	0.727	FALSE	—
1	EOJ	Ensemble vs Control	65	1	FALSE	—
2	EOJ (Majority Vote)	Humans vs Control	65	1	FALSE	—
2	EOJ (Majority Vote)	Scenario Forecasting vs Control	65	1	FALSE	—
2	EOJ (Majority Vote)	Prompt Engineering vs Control	65	0.688	FALSE	—
2	EOJ (Majority Vote)	Most Popular Option vs Control	65	1	FALSE	—
2	EOJ (Majority Vote)	Delphi vs Control	65	0.688	FALSE	—
2	EOJ (Majority Vote)	Ensemble vs Control	65	1	FALSE	—
2	EOJ (Majority Vote)	Meta vs Control	65	1	FALSE	—

Table 4 Combined confusion matrix comparing number and percent of correct and incorrect Scenario Forecasting and Prompt Engineering predictions with those of the Control.

		Scenario Forecasting		Prompt Engineering	
		Correct	Incorrect	Correct	Incorrect
Control	Correct	339 (49.1%)	36 (5.2%)	325 (47.1%)	50 (7.2%)
	Incorrect	65 (9.4%)	250 (36.2%)	77 (11.2%)	238 (34.5%)

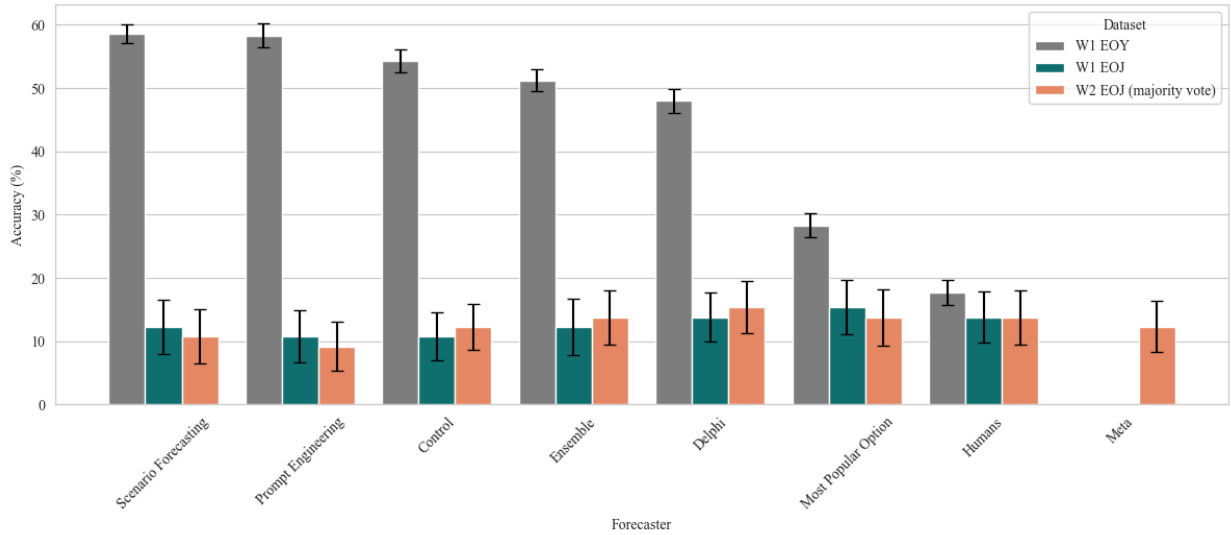

Figure 2 Forecasting accuracy by model and dataset visualizes the significant drop in accuracy from the EOY dataset to the EOJ dataset. Error bars (± 1 standard error) show that the small size of the EOJ dataset contributed to greater uncertainty in the estimate of the proportion correct.

Prediction accuracy for the EOJ events was uniformly low across all models and waves (see Figure 2). Across Wave 1 and Wave 2 predictions, no model exceeded 15.4% accuracy on the EOJ dataset. Human forecasts similarly performed poorly on the EOJ dataset, never exceeding 13.8% accuracy, which was not significantly different from any model in EOJ (see Table 1). Across both EOJ waves, no statistically significant differences in accuracy were observed between the Human benchmark, the Control model, or any of the other forecasting approaches (all McNemar tests, $p > 0.05$) (see Tables 2 and 3). Furthermore, no significant

differences were observed between Wave 1 and Wave 2 predictions for EOJ events (McNemar tests, all $p = 1.00$).

In contrast, prediction accuracy was substantially higher for the end-of-year (EOY) dataset (see Figure 2). Across models, EOY accuracy averaged 49.8%, with individual model accuracy ranging from 28.3% to 58.6% (see Table 1). Humans performed markedly worse on the EOY dataset, achieving 17.7% accuracy, which was significantly lower than every other EOY model tested (Exact McNemar tests, all $p < 0.05$) (see Table 2).

Among the EOY models, Scenario Forecasting, with 58.6% accuracy, and Prompt Engineering, with 58.3% accuracy, achieved the highest accuracy scores, both significantly outperforming the Control model with 54.3% accuracy (Exact McNemar tests, $p < 0.05$). The confusion matrix comparing Scenario Forecasting and Prompt Engineering with the Control model shows that both methods produced more correct predictions when the Control was incorrect, as Scenario Forecasting correctly predicted 65 events that the Control predicted incorrectly and Prompt Engineering correctly predicted 77 such events (see Table 4). Delphi (48.0%) and Ensemble (51.2%) performed moderately well but did not exceed the Control model (see Table 3). The Most Popular Option model performed substantially worse than the other models, achieving only 28.3% accuracy. Notably, 78.8% of Most Popular Option predictions matched the Human prediction, whereas only 42.0% of Control model predictions matched the Human prediction.

Finally, accuracy declined significantly from the EOY dataset to the EOJ dataset across all models (two-proportion z-tests, all $p < 0.05$). The only exception was the Human forecasts, for which the difference between EOY and EOJ accuracy was not statistically significant.

DISCUSSION

The results of this study differed substantially across datasets. Forecasting accuracy for the EOJ events was distinctly low across all models and waves, with no statistically significant differences observed between models or between waves. This suggests that forecasting three separate passes of the EOJ dataset and then applying majority-vote aggregation did not meaningfully improve performance on these questions, as the Wave 1 EOJ dataset was predicted once, as opposed to the Wave 2 EOJ dataset, on which each model made three passes. One possible explanation for the low accuracy on the EOJ dataset is that EOJ events were inherently more difficult to forecast as a result of their date of resolution. The EOY dataset contained only events that were set to expire at the end of 2025. These questions, therefore, were likely more prevalent in the models' online resources than questions that would resolve at a more arbitrary date, such as the end of January. In addition, this unpopularity likely caused the EOJ dataset's relatively small sample size (n = 65), which was a probable factor in the low accuracy scores, as AI systems often require large datasets to be effective [16]. As demonstrated by the error bars in Figure 2, the small size of the EOJ dataset meant that there was greater uncertainty in the observed accuracies as estimates of the models' overall forecasting performance.

Human forecasting accuracy was also low relative to model performance, particularly on the EOY dataset. The Human benchmark, represented by Kalshi's most popular human prediction for each event, achieved only 17.7% accuracy for EOY events and did not perform significantly better than any model for EOJ events. Because this benchmark reflects the most popular market position rather than directly collected expert predictions, it may not fully capture human forecasting ability. Open interest can reflect other factors such as trading strategies, hedging (using event contracts as one would use insurance, reducing real-world financial risk by

betting on positions opposite one's circumstance), low engagement with certain markets, or automated trading bots. As a result, it should be interpreted more as a baseline based on market trends than as a measure of human forecasting performance.

Among the EOY models, Scenario Forecasting and Prompt Engineering had the highest predictive accuracy, and both significantly outperformed the Control model. These results suggest that prompts that can organize reasoning and aid in evaluating possible outcomes have the ability to improve LLM forecasts. By forcing the model to consider optimistic, pessimistic, and typical outcomes or to follow a structured reasoning process, these approaches may reduce common reasoning biases such as anchoring on the first plausible answer or on surface-level data [17]. In addition, the engineered prompts were themselves generated by an LLM, demonstrating that AI systems can construct their own pre-prompts to improve predictive accuracy. This suggests that in real-world applications, the prompt design process could be automated from a single prompt, saving researchers time and effort. More broadly, these results indicate that frameworks designed to enhance structured reasoning in LLMs such as the aforementioned ART (Automatic Reasoning and Tool-use), originally developed to help models solve reasoning tasks in the present, may also be useful for forecasting future events by guiding models through more systematic evaluation of possible outcomes.

In contrast, the automated Delphi procedure and the Ensemble method did not outperform the Control model. When a model is shown multiple predictions generated by a version of itself, like in the Ensemble, Delphi, or Meta models, it may simply reinforce any initial reasoning patterns instead of encouraging diverse modes of prediction [3]. This tendency to adopt bias is also shown in the Most Popular Option model. This model performed substantially worse than the other models. Its low accuracy is consistent with the fact that its

predictions closely mirror the market's modal predictions (78.8% similarity), which also performed poorly relative to the models. The strong overlap between Most Popular Option predictions and the Human benchmark indicates that this method very often reproduces the same output as its input. From this evidence that AI forecasters tend to cling to given information, a more effective alternative, illustrated by the successful Scenario Forecasting and Prompt Engineering models, may be to guide the model on how to reason about a problem rather than exposing it to solutions.

This study has several limitations. All forecasts were generated using a single Western-developed LLM and questions drawn from the U.S.-based Kalshi prediction market, which may have presented cultural and informational biases in both the model's training data and the market behavior, potentially restricting the generalizability of these results to U.S. contexts. If forecasting systems like those tested in this study were eventually used to influence large-scale decisions, their potentially biased outputs could come to carry significant authority in how the public forms expectations about future events, begging the question of how much influence automated predictions should have over human judgment. In terms of statistical analysis, the EOJ dataset was smaller and more arbitrarily timed than the EOY dataset, causing the statistical tests to detect smaller performance differences between models and limiting online resources for the models to review. In addition, Human forecasts were approximated using market open interest rather than being directly collected, which may not accurately reflect the performance of motivated or expert forecasters. Future research should test these forecasting methods on larger datasets containing events with limited historical data to resimulate the EOJ prediction and better understand how AI models perform in environments where prior information is sparse. Evaluating models—like those used in this study and mentioned in the introduction—on

quantitative prediction markets may also allow for the use of more analytically robust strategies that could aid judgmental forecasts when historical data is unavailable.

CONCLUSION

This study evaluated several forecasting strategies to determine whether prompting- and judgment-based techniques can improve AI prediction accuracy. The results showed that Scenario Forecasting and Prompt Engineering significantly improved model performance on end-of-year events, while the Control model outperformed more complex aggregation methods such as Delphi and Ensemble on the same dataset. Prediction accuracy declined sharply for end-of-January events. While this decline can be primarily attributed to the EOJ dataset's small sample size, it could also suggest that limited online reference material may reduce the effectiveness of AI forecasts. In addition, the findings reveal that guiding models on how to reason about a problem may be more effective than exposing them to existing predictions or solutions, as AI forecasters were found to be prone to quickly adopting biases by clinging to provided information. In real-world applications, this study suggests that LLMs could partially automate prompt design, allowing researchers to quickly generate case-specific prompts and improve AI output with minimal manual effort. More broadly, these results indicate that reasoning frameworks like Automatic Reasoning and Tool-use may be useful not only for solving present reasoning tasks but also for forecasting future events by guiding models through more systematic evaluation of possible outcomes. Overall, the findings of this study suggest that structured reasoning prompts can enhance AI forecasting performance in data-rich environments while introducing external perspectives creates an overly biased model, and additional research

is needed to better understand how these methods perform when online reference material is scarce.

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